Alibaba Cloud

Promo-proof checkout in Jakarta

Architecture proposal

Prepared for Segar Grocer

Date	Revisions	Products	Catalog checked
2 November 2026	1	12	2026-10-31

Prepared with Architecture Builder

AI-generated. Review with a solutions architect before you build.

Executive summary

Segar Grocer runs an online grocery in Jakarta where checkout crashes during weekend promos drive customers to abandon carts.

This proposal designs a promo-proof checkout on Alibaba Cloud, hosted entirely in Jakarta, that scales automatically and keeps carts safe when traffic spikes.

The result is a checkout that stays fast under load, protecting weekend revenue and keeping shoppers from leaving.

What this design delivers

Revenue protected during weekend promos	Auto Scaling and order queuing keep checkout responsive when traffic surges, so shoppers complete purchases instead of abandoning carts.
Carts survive application restarts	Tair holds shopping carts in memory across zones, so a server restart during a promo never empties a customer's basket.
Faster pages for Jakarta shoppers	ESA caches product images at the edge and accelerates dynamic checkout calls, reducing wait times that lead to abandonment.
Bots blocked during high-demand promos	WAF filters out scalper bots and injection attacks, reserving inventory and bandwidth for real customers.

What you told us, and how the design answers it

Each priority from our conversation, where it came from, and the part of the design that answers it.

Checkout crashes during weekend promos cause cart abandonment.
“checkout crashes during weekend promos”

How the design answers it
Auto Scaling adds servers before the crash point. RocketMQ buffers orders so payment delays never stall the page. Shoppers finish buying even at peak load.

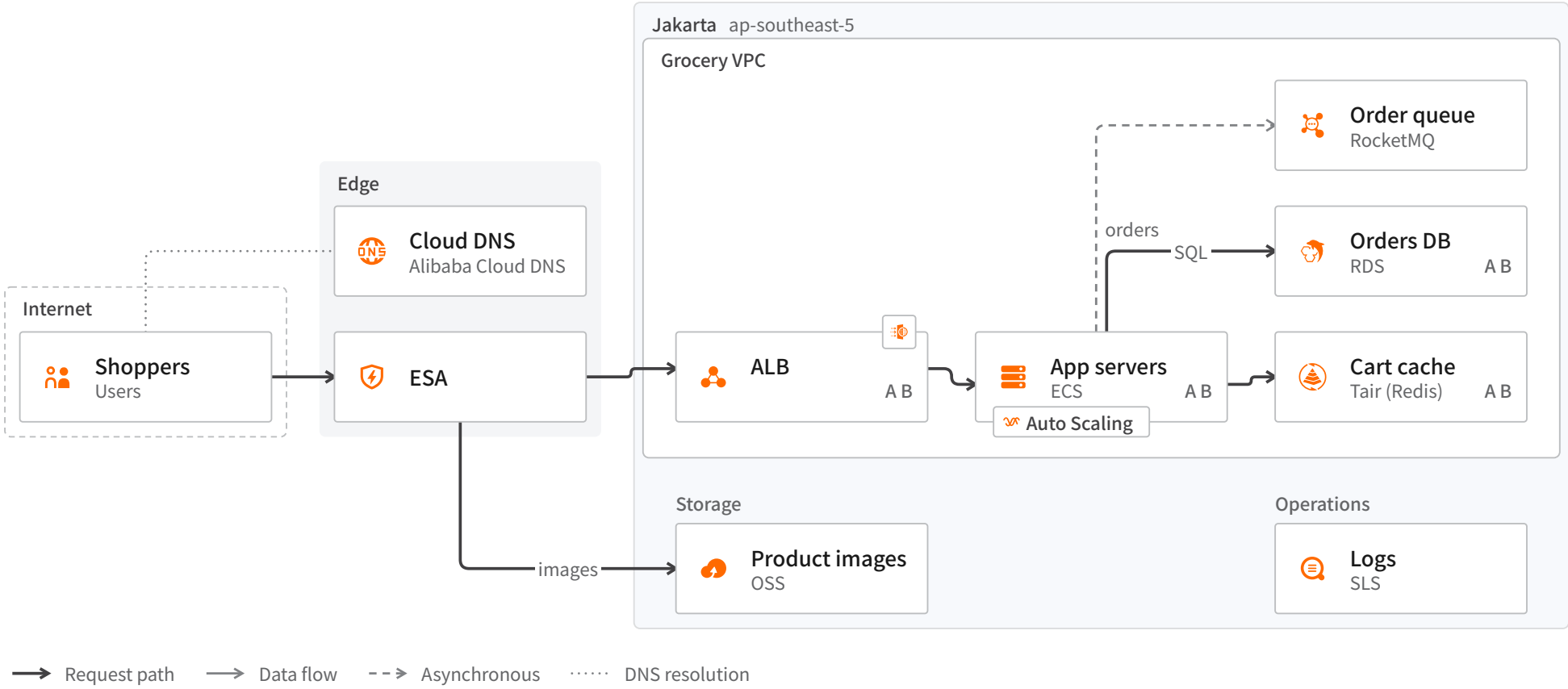
Products involved
Auto Scaling (ESS), ApsaraMQ for RocketMQ

All workloads and data must remain in Jakarta.
“We run an online grocery in Jakarta”

How the design answers it
Every component in this design runs in the ap-southeast-5 region. This keeps data local and avoids cross-border latency for your shoppers.

Products involved
Elastic Compute Service (ECS), ApsaraDB RDS, Tair (Redis® OSS-Compatible), Object Storage Service (OSS), Simple Log Service (SLS), Application Load Balancer (ALB), ApsaraMQ for RocketMQ

Architecture diagram



Components

Every product in the diagram, why it is in this design, and where to read its official documentation.

Users

Shoppers
Internet

Why it's here
Customers browsing and checking out groceries.

What it does
People reaching the system over the internet from browsers or apps.

Official documentation
Documentation link not yet available

Alibaba Cloud DNS

Edge · Domain Names and Websites

Why it's here
Shoppers need to reach the store reliably. DNS resolves your domain to the edge network quickly. Without it, customers cannot find the site at all.

What it does
Authoritative public DNS for your domains; a resolve hop, not on the data path.

Official documentation
www.alibabacloud.com/product/dns

Edge Security Acceleration (ESA)

Edge · Networking & CDN

Why it's here
Slow pages drive cart abandonment. ESA serves cached grocery photos from the edge and speeds up checkout API calls. Without it, every image request hits Jakarta, slowing the experience.

What it does
Global edge: caching, dynamic acceleration, WAF, DDoS defense, bot management and edge functions.

Official documentation
www.alibabacloud.com/product/esa

Application Load Balancer (ALB)

Jakarta (ap-southeast-5) · Grocery VPC · Zones A, B

Why it's here
Traffic must spread evenly so no single server fails under promo load. ALB routes requests across two availability zones. Without it, one zone failure could take down checkout.

What it does
Layer-7 HTTP/HTTPS/gRPC load balancer with content routing; also serves as ACK Ingress.

Official documentation
www.alibabacloud.com/product/server-load-balancer

Web Application Firewall (WAF)

Jakarta (ap-southeast-5) · Grocery VPC · On ALB

Why it's here

Promos attract bot scalpers that steal stock and slow the site. WAF blocks these attacks at the load balancer. Without it, fake traffic competes with real shoppers.

What it does

WAF 3.0: blocks OWASP attacks, CC floods and bots via cloud-native or CNAME access.

Official documentation

www.alibabacloud.com/product/web-application-firewall

Elastic Compute Service (ECS)

App servers

Jakarta (ap-southeast-5) · Grocery VPC · Zones A, B

Why it's here

Your storefront and checkout logic need compute to run. ECS hosts the application in two zones. Without it, there is no platform to serve customers.

What it does

Virtual machines (x86, Arm, GPU) with cloud disks and security groups.

Official documentation

www.alibabacloud.com/product/ecs

Auto Scaling (ESS)

Jakarta (ap-southeast-5) · Grocery VPC · On App servers

Why it's here

Weekend promos spike traffic beyond fixed capacity. ESS adds ECS instances automatically when CPU rises. Without it, you either overpay for idle servers or crash under load.

What it does

Scales ECS or ECI fleets by rules, schedules or metrics and registers them to load balancers.

Official documentation

www.alibabacloud.com/product/auto-scaling

ApsaraMQ for RocketMQ

Order queue

Jakarta (ap-southeast-5) · Grocery VPC · Middleware

Why it's here

A slow payment gateway can block the entire checkout. RocketMQ buffers order events so the page stays fast. Without it, payment delays directly cause cart abandonment.

What it does

Managed RocketMQ 5.x for transactional, scheduled and ordered messaging.

Official documentation

www.alibabacloud.com/product/mq

ApsaraDB RDS

Orders DB

Jakarta (ap-southeast-5) · Grocery VPC · Zones A, B

Why it's here

Orders and inventory must persist accurately. RDS stores this data with zone-level high availability. Without it, a database outage halts all sales.

What it does

Managed MySQL, PostgreSQL, SQL Server and MariaDB with HA, read replicas and backups.

Official documentation

www.alibabacloud.com/product/apsaradb-rds

Tair (Redis® OSS-Compatible)

Cart cache

Jakarta (ap-southeast-5) · Grocery VPC · Zones A, B

Why it's here

Losing a cart means losing a sale. Tair keeps carts in memory across zones so they survive app restarts. Without it, any crash forces shoppers to start over.

What it does

Managed Redis-compatible in-memory database and cache, plus Tair enterprise features.

Official documentation

www.alibabacloud.com/product/tair

Object Storage Service (OSS)

Product images

Jakarta (ap-southeast-5) · regional services · Storage

Why it's here

Grocery photos need durable storage that ESA can pull from quickly. OSS holds these assets in Jakarta. Without it, images have no reliable source.

What it does

Durable object storage for files, media, backups and data lakes; S3-compatible.

Official documentation

www.alibabacloud.com/product/object-storage-service

Simple Log Service (SLS)

Logs

Jakarta (ap-southeast-5) · regional services · Storage

Why it's here

When checkout slows, teams need logs to find the cause fast. SLS collects application and checkout logs. Without it, diagnosing a crash takes longer and costs more sales.

What it does

Collects, searches, analyzes with SQL and alerts on logs, metrics and traces.

Official documentation

www.alibabacloud.com/product/log-service

Considerations and next steps

Trade-offs and how they are handled

The design assumes weekend promo spikes up to 10x normal traffic. If actual spikes are larger, scaling limits may be reached.

How it is handled

Confirm expected concurrent shoppers during peak promos to size Auto Scaling rules correctly.

RocketMQ decouples order placement from payment processing, but the payment integration method is not yet defined.

How it is handled

Confirm whether payment uses a third-party gateway or a built-in service to finalize the queue configuration.

Next steps

1. Confirm expected concurrent shoppers during peak weekend promos: Under 10k, 10k to 50k, or Over 50k.
2. Confirm how payment is processed today: Third-party payment gateway or Built-in payment service.
3. Review the catalog pricing in the attached PDF to validate the budget for weekend scaling.

Design record

What the design rests on: the requirements pinned in the conversation, constraints, assumptions and open questions, with the notes from the first drawing.

The design

I'll design a Jakarta grocery checkout that autoscales on weekends, with caching and a resilient database to stop cart abandonment. ECS instances run the app and scale automatically for weekend promos. Tair holds carts so they survive crashes. RocketMQ decouples order placement from payment processing so a slow gateway never blocks checkout.

12 products, 9 connections, 1 revisions

Pinned requirements

Workloads stay in Jakarta

Held
Pinned in revision 1

Constraints

Data and workloads stay in Jakarta (ap-southeast-5)

Assumptions

Weekend promo spikes up to 10x normal traffic
Alternative: Steady traffic without large spikes

Public-facing grocery store
Alternative: Internal staff portal only

Open questions

How many concurrent shoppers do you expect during peak weekend promos? (Under 10k / 10k to 50k / Over 50k)

How is payment processed today? (Third-party payment gateway / Built-in payment service)

Connections

How the products in the diagram talk to each other.

From	To	Carries	Type
Shoppers	Cloud DNS		DNS resolution
Shoppers	ESA	HTTPS	Data
ESA	ALB	origin	Data
ALB	App servers		Data
App servers	Cart cache		Data
App servers	Orders DB	SQL	Data
App servers	Order queue	orders	Asynchronous
ESA	Product images	images	Data
App servers	Logs		Control

Revision history

Every ask and what it changed in the design.

Revision 1

Drawn in 23.7 s

Ask

We run an online grocery in Jakarta. Our checkout crashes during weekend promos and customers abandon carts. Design it on Alibaba Cloud.

Changes

+ Users · Shoppers · in Internet
+ Cloud DNS
+ ESA · in Edge
+ ALB · zones A, B
+ WAF · on ALB
+ ECS · App servers · zones A, B
+ Auto Scaling · on ECS
+ Tair (Redis) · Cart cache · zones A, B
+ RDS · Orders DB · zones A, B
+ OSS · Product images · in Jakarta (ap-southeast-5)
+ RocketMQ · Order queue · in Grocery VPC
+ SLS · Logs · in Jakarta (ap-southeast-5)
+ 9 connections